

CHyLL v2: Continuous Latent Embeddings of Hybrid Systems

Method, loss terms, and a guide to the trained models and data

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Hybrid Dynamical Systems

A *hybrid* dynamical system mixes smooth motion with sudden jumps.

- A passive walker: the swing leg moves smoothly, then strikes the ground and the state jumps.
- A bouncing ball: free flight under gravity, then an instantaneous change of velocity at each impact.

The jump is a genuine discontinuity. A smooth model of the dynamics cannot be fitted directly across it.

Three Example Systems

System	State dim.	Behavior
Rimless wheel	2	stable rolling limit cycle
Bouncing ball	2	bounces, losing energy at each impact
Compass-gait biped	4	period-doubling route to chaos

They increase in difficulty. The compass-gait biped walks with period 1, then 2, 4, 8, and finally chaotically as the ground slope steepens.

The Idea

Replace the discontinuous system with a continuous one that is easy to model.

- An **encoder** maps each state into a latent space.
- A single **smooth vector field** advances the latent state in time.
- A **decoder** maps the latent state back.

The jump is absorbed into the smooth latent flow. This follows Teng, Liu, and Sreenath (ICML 2026).

The Mapping-Cylinder State Space

To make the jump continuous, the state is augmented with one extra coordinate $s \in [0, 1]$:

$$y = (x, s) \in X'.$$

- $s = 0$: ordinary motion of the original system.
- s sweeping $0 \rightarrow 1$: the jump, unrolled as a continuous path.

On X' the trajectory is continuous, so a smooth latent model can be fitted to it.

The Learned Maps

CHyLL v2 learns three maps. Write $Z = \mathbb{R}^d$ for the latent space.

$E_\theta : X' \rightarrow Z$	encoder,
$V_\theta : Z \rightarrow Z$	latent vector field,
$D_\theta : Z \rightarrow X'$	decoder.

The time- τ flow of V_θ is written F_θ . The latent dimension is $d = 7$ for the rimless wheel and bouncing ball, and $d = 11$ for the compass-gait biped.

Training: The Forward Pass

For a sampled trajectory slice $y_0, \dots, y_{T-1}$:

$$z_i = E_\theta(y_i),$$

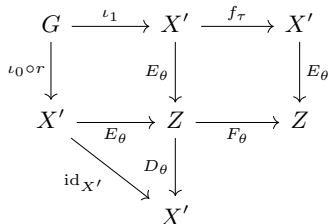
$$\hat{z}_i = F_\theta^i(z_0),$$

$$\hat{y}_i = D_\theta(\hat{z}_i).$$

Each state is encoded, the latent flow is integrated forward, and the integrated latent trajectory is decoded. The loss compares the encoded z_i , the integrated $\hat{z}_i$, and the decoded $\hat{y}_i$ against the data.

The Commutative Diagram

Here f_τ is one time- τ step of the system, G the impact set, r the reset map, and ι_0, ι_1 the inclusions at $s = 0$ and $s = 1$. The learned maps should make this diagram commute:



Each cell is enforced by one loss term:

- Left square, L_g : E_θ sends both sides of a reset to one latent point.
- Top-right square, L_z : the latent flow matches the dynamics.
- Lower triangle, L_x : the decoder inverts the encoder.

The Diagram Loss Terms

Write $z_i = E_\theta(y_i)$ and $\hat{z}_i = F_\theta^i(z_0)$. The sampled data satisfies $y_{i+1} = f_\tau(y_i)$.

Latent consistency. $L_z = \frac{1}{T} \sum_i \|\hat{z}_i - E_\theta(y_i)\|^2$

$L_z = 0 \Rightarrow F_\theta \circ E_\theta = E_\theta \circ f_\tau$, the semi-conjugacy square.

Reconstruction. $L_x = \frac{1}{T} \sum_i \|D_\theta(\hat{z}_i) - y_i\|^2$

$L_z = L_x = 0 \Rightarrow D_\theta \circ E_\theta = \text{id}_{X'}$, the reconstruction triangle.

Gluing. $L_g = \frac{1}{|\mathcal{G}|} \sum_{k \in \mathcal{G}} \|E_\theta(y_{k+1}) - E_\theta(y_k)\|^2$

$L_g = 0 \Rightarrow E_\theta \circ \iota_1 = E_\theta \circ \iota_0 \circ r$ on G , the gluing square.

The Regularity Loss Terms

L_v and L_c are not diagram cells. They keep the embedding well-behaved.

Seam velocity. The seam set $\mathcal{S}$ is the union of the cylinder-entry seams and the gluing-exit seams. With $z_k = E_\theta(y_k)$ and step τ ,

$$L_v = \frac{1}{|\mathcal{S}|} \sum_{k \in \mathcal{S}} \left\| \frac{z_k - z_{k-1}}{\tau} - \frac{z_{k+2} - z_{k+1}}{\tau} \right\|^2.$$

Minimizing L_v brings the latent velocity just before and just after each seam into agreement, the C^1 (no-kink) condition the loss aims to enforce.

Anti-collapse. With σ_j^2 the variance of latent coordinate j and floor λ , $L_c = \sum_{j=1}^d \max(0, \lambda - \sigma_j^2)$. Any coordinate with variance below λ is penalized, so E_θ keeps all d coordinates active.

The edge sets $\mathcal{G}$ and $\mathcal{S}$ are detected from the sampled coordinate s , not from symbolic guard equations.

The Composite Objective

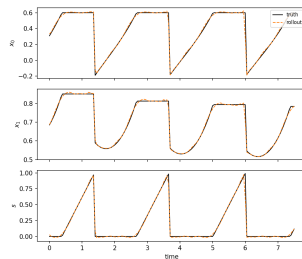
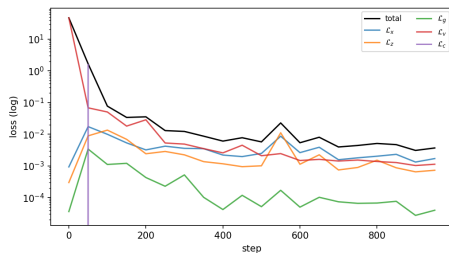
$$L = w_x L_x + w_z L_z + w_g L_g + w_v L_v + w_c L_c.$$

Setting	$(w_x, w_z, w_g, w_v, w_c)$
Rimless wheel, bouncing ball	(1, 1, 3, 1, 1)
Compass-gait biped	(1, 1, 3, 0, 1)

The exact values for any run are stored in its `config.json`.

Training: Rimless Wheel

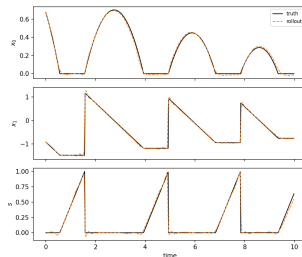
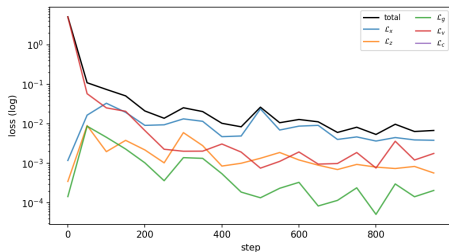
Training has two phases: Phase A from a random start, then Phase B fine-tuning with the seam-velocity term L_v . The loss falls smoothly across the curriculum and the rollout tracks the true trajectory through many impacts.



Left: loss history. Right: decoded rollout against ground truth. Final composite loss $\approx 3.7 \times 10^{-3}$.

Training: Bouncing Ball

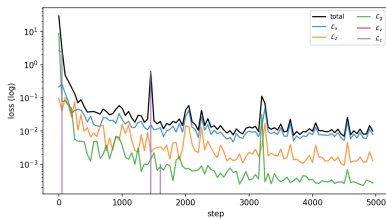
The bouncing ball uses the same two-phase training. Each impact removes energy, so the true trajectory is a decaying sequence of bounces, which the rollout reproduces.



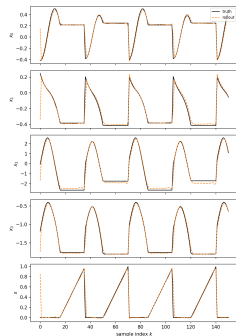
Left: loss history. Right: decoded rollout against ground truth. Final composite loss $\approx 6.8 \times 10^{-3}$.

Training: Compass-Gait Biped

The compass-gait biped trains in a single phase, one run per ground slope. Shown is the period-2 gait at 4.75° ; the steeper cascade slopes train the same way.



loss history



rollout vs. truth

Final composite loss $\approx 1.0 \times 10^{-2}$. The rollout panel shows the augmented state, four physical variables and the coordinate s .

Results

- **Rimless wheel.** The model reproduces the rolling limit cycle, and the learned embedding recovers its expected single topological cycle.
- **Compass-gait biped.** The cascade analysis recovers the period-doubling sequence exactly: the learned latent states form 2, 4, and 8 clusters at the period-2, -4, and -8 gaits, and disperse under chaos.
- **Bouncing ball.** The orbit spirals slowly toward rest rather than closing into a loop, so its topological signature is only partially recovered.

Where the Models and Data Live

Path	Contents
<code>runs/<name>/</code>	trained model, config, training log
<code>figures/<name>/</code>	loss and rollout figures
<code>cycling_signature/data/</code>	exported lifts and rank summaries
<code>cycling_signature/figures/</code>	rank distribution and heatmap figures
<code>compass_analysis/</code>	compass cascade cluster analysis

Paths are under `chyll_v2/`. Each `<name>` is one trained run, for example `rimless_wheel_phaseB_finetune`.

A Trained Run

Each directory in `chyll_v2/runs/` contains:

- `model.pt`: the learned network weights.
- `config.json`: dimensions, system parameters, and loss weights.
- `train_log.jsonl`: the loss terms recorded over training.

The config is enough to rebuild the network and reload the weights. The export scripts under `cycling_signature/export/` do exactly this.

What an Exported Lift Contains

An exported lift is a sampled trajectory in the latent space Z :

- `..._positions`: the sampled latent points, one per row.
- `..._tangents`: the unit tangent direction at each point.

Both arrays have shape (N, d) : N samples and d latent coordinates. Each lift is written as `.npy` for Python and `.csv` for inspection.

Loading a Lift in Python

```
import numpy as np

folder = "chyll_v2/cycling_signature/data/rimless_wheel"
base = folder + "/continuous_lift_chyll_v2_phaseB"

Z = np.load(base + ".npz")          # positions
T = np.load(base + "_tangents.npz") # unit tangents

print(Z.shape, T.shape)
print(np.linalg.norm(T, axis=1).max())
```

Z holds the latent positions and T the unit tangents. The printed tangent norm should be close to 1.

From a Trained Model to a Rank Figure

The topology analysis runs in three stages:

- 1 **Export**: a script in `cycling_signature/export/` turns a trained model into a latent lift.
- 2 **Compute**: `cycling_signature/julia/run_subsegments.jl` computes the cycling signature over random subsegments.
- 3 **Plot**: `cycling_signature/plot/plot_signature_figures.py` draws the rank distribution and heatmaps.

The `cycling_signature/` folder has its own README with the exact commands.

Sanity Checks

After loading a lift:

- positions and tangents have the same shape,
- tangent norms are close to 1,
- the latent dimension d matches the system, 7 or 11.

A common mistake is pairing a model with the wrong `config.json`: the network shape will not match the saved weights. Take `model.pt` and `config.json` from the same run directory.

Where the Code Lives

Object	File
$E_\theta, V_\theta, D_\theta$	<code>networks.py</code>
latent ODE rollout	<code>ode.py</code>
loss terms L_x, L_z, L_g, L_v, L_c	<code>losses.py</code>
training loop	<code>train.py</code>
system simulators and reset maps	<code>systems/</code>

These files are the CHyLL v2 core library, under `chyll_v2/chyll_v2/`. Topology postprocessing lives under `chyll_v2/cycling_signature/`.

Summary

- A hybrid system is made continuous on the mapping cylinder X' .
- CHyLL v2 learns an encoder, a latent vector field, and a decoder.
- L_g , L_z , and L_x are the commuting cells of the diagram.
- L_v and L_c add the seam-regularity conditions.
- The learned embeddings recover the expected topology for the rimless wheel and the compass-gait cascade.